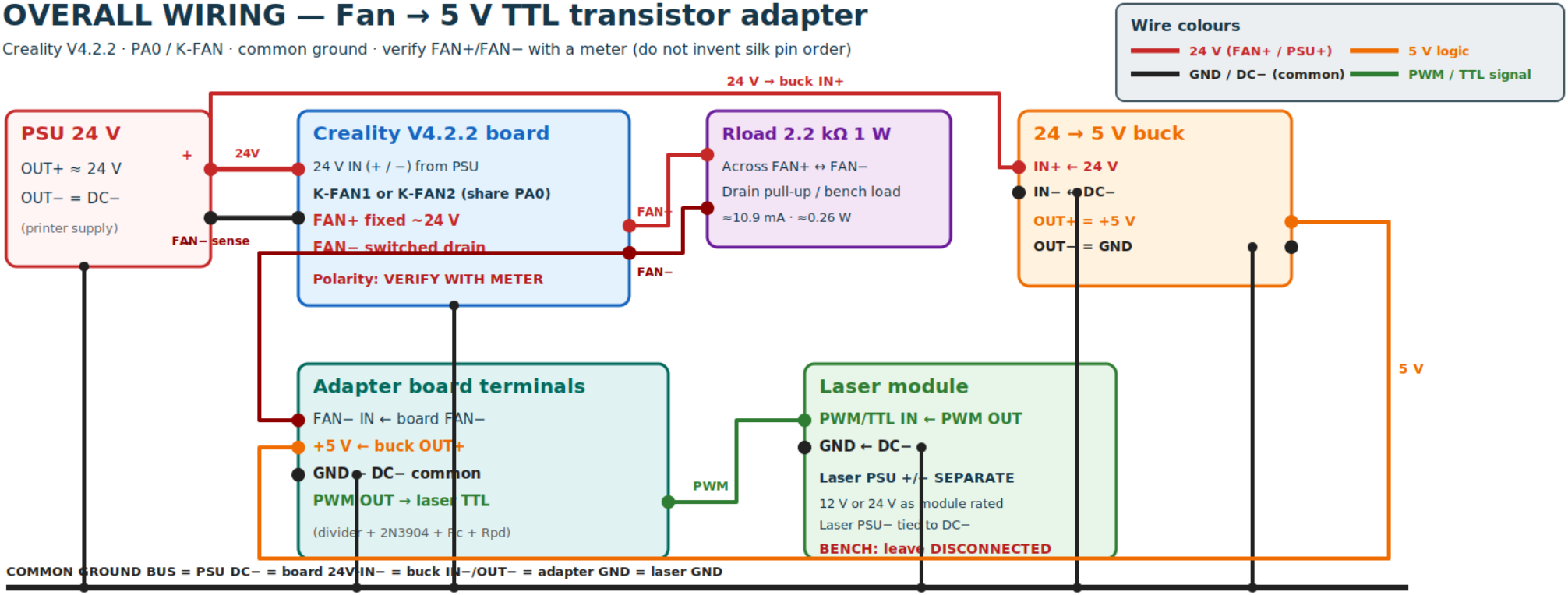
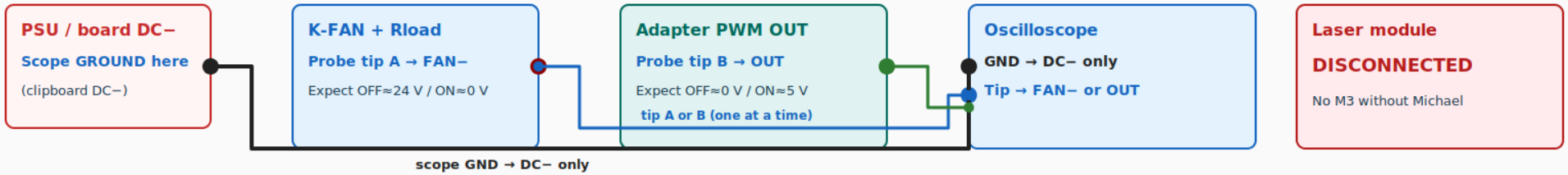
OVERALL WIRING — Fan → 5 V TTL transistor adapter

Creality V4.2.2 · PA0 / K-FAN · common ground · verify FAN+/FAN− with a meter (do not invent silk pin order)



BENCH TEST VARIANT — laser disconnected · Rload only · heaters off

Use SCOPE8 Fan PWM or SCOPE9 FanPA0 menus. Scope ground ONLY on DC−. Never clip ground to FAN−.



Expected with Rload (tip vs DC−)

Fan GPIO/PWM OFF: FAN− ≈ 24 V OUT ≈ 0 V

Fan GPIO/PWM ON: FAN− ≈ 0 V OUT ≈ 5 V

PWM 25/50/75/100%: FAN− inverted power PWM @ 1 kHz; OUT upright 0/5 V same duty

Never ground-clip FAN− or BED−. Heaters disconnected. Laser disconnected for this bench.